

"PAPER_{0:D3}" -f [int]# PAPER #75 ■ X-Ray Binaries: Chandra + UQFF Field Analysis

Author: Daniel T. Murphy

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<!-- UQFF constants: $\tau = 5.0\text{e-}4 \text{ day}$ ■, $[\text{SSq}] = 0.57$, $M_{\text{UQFF}} = 1.43\text{e}1 \text{ TeV}$ -->

Abstract

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2. UQFF XRB Luminosity Model

Standard Accretion Efficiency

$$\eta_{\rm Eddington} = 0.1 \times \frac{L_X}{L_{\rm Edd}}$$

UQFF-Enhanced Efficiency

$$\eta_{\rm UQFF} = \eta_{\rm Edd} \times (1 + [SCm]) = \eta_{\rm Edd} \times 1.99$$

Where [SCm] = 0.99 (superconductive vacuum coupling, Batch 23).

This UQFF enhancement predicts X-ray luminosities ~2x higher than the Eddington limit in strongly magnetized systems consistent with **ultra-luminous X-ray sources (ULX)** observed by Chandra.

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X-1 ULX (NGC 5907)	NS-ULX	17,000	1.0x10 ⁴⁴	2.0x10 [?]	4.0x10 [?]	25x

The NGC 5907 X-1 ULX line shows that even the UQFF 2x enhancement cannot fully explain super-Eddington ULX emission these systems require additional geometric beaming or magnetic field confinement beyond the basic UQFF Superconductive mode.

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$$\Delta HR_{\rm UQFF} = [UA] \times \frac{n_{\rm vac}}{n_{\rm ISM}} \times HR_{\rm standard} = 0.0001 \times 10^{-6} \times HR = \text{negligible}$$

Result: HEASARC hardness ratios are unmodified by UQFF at measurable precision. UQFF modifies luminosities (via ?_UQFF), not spectral shape.

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# Cone search: 1  $\times$  radius around Cygnus X-1
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params = {
'ra': 299.590, 'dec': 35.2016,
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$\eta_{\rm Eddington} = 0.1 \times \frac{L_X}{L_{\rm Edd}}$

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Where [SCm] = 0.99 (superconductive vacuum coupling, Batch 23).

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GRS 1915+105	BH	8.6	6.0x10 ³⁸	7.4x10 ³⁸	7.5x10 ³⁸	0.80
X-1 ULX (NGC 5907)	NS-ULX	17,000	1.0x10 ⁴⁴	2.0x10 [?]	4.0x10 [?]	25x

The NGC 5907 X-1 ULX line shows that even the UQFF 2x enhancement cannot fully explain super-Eddington ULX emission = these systems require additional geometric beaming or magnetic field confinement beyond the basic UQFF Superconductive mode.

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The HEASARC XRAYBSC catalog provides 235 bright X-ray sources detected by ROSAT.

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Accretion efficiency	10%	~20% ([SCm]■Edd)	Compatible with ULX
Hardness ratio	Standard	+ [UA] correction (negligible)	Unmodified
ULX luminosity	1■10■ Edd	2■ Edd + beaming	Requires beaming
Typical XRB L_X	Eddington	■15■25%	< 2s agreement

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Appendix: UQFF Production Framework Reference (v4.75+)

*Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references
the production physics constants and master equations to enable reproducibility
against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-11} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k█	1.5	Ug1 DPM-dipole coupling
k█	1.2	Ug2 outer-bubble charge coupling
k█	1.8	Ug3 string-rotation coupling
k█	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale

Symbol	Value	Description
E_react(0)	10 J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 \lambda_i \int d^3x \, U_i(r,t) \cdot E_{\mathrm{react}} \cdot U_i(r,t)$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\sum \lambda_i \cdot U_i \cdot E_{\mathrm{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420): $\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\mathrm{full}} = U_m^{\mathrm{base}} \cdot \exp\left(\frac{1}{1 + 10^{13} \cdot \left(\frac{ho_{SCm}}{ho_c}\right) \cdot \cos(\Delta\omega \cdot t)}\right)$$

Symbol	Value	Description
ρ_c	10^1 kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.